

Compound Word Transformer: Learning to Compose Full-Song Music over Dynamic Directed Hypergraphs

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Motivation

- Transformer as a strong music generation model
 - Pop music
 - mean: 6432 tokens
 - **max** $\geq$ **10K** tokens
 - memory complexity
 - vanilla transformer: $O(N^2)$
- Crop one song into segments
- Can we generate a music piece of full-song length?

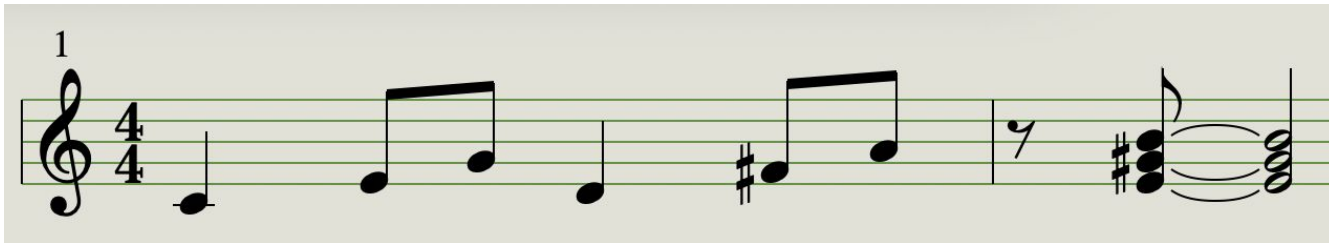
Motivation

- Reduce the length of the token sequence
 - Novel compact representation of music
 - Compound Word (**CP**)
- Lower Memory Complexity
 - Advanced transformer
 - transformer-XL
 - linear transformer

Overview

- Representation: From MIDI to CP
- Model
- Experiments
 - Tasks
 - Unconditional Generation
 - Conditional Generation
 - Evaluation
 - Quantitative Evaluation
 - Qualitative Evaluation

Representation: MIDI



...

```
<note_on, note=60, velocity=94, time=0.0 sec>
```

```
<note_on, note=60, velocity=0, time=1.253 sec>
```

```
<note_on, note=64, velocity=94, time=1.253 sec>
```

```
<note_on, note=64, velocity=0, time=1.879 sec>
```

```
<note_on, note=67, velocity=94, time=1.879 sec>
```

```
<note_on, note=67, velocity=0,  time=2.506 sec>
```

...

Sequence Length: 18
9 (notes) x 2 (note on/off)

...

```
pitch: 60,  
velocity:94,  
time: 1.20,
```

...

Sequence Length: 54
9 (notes) x 2 (note on/off) x 3 (attributes)

Representation: MIDI

- Problem
 - 1 note, 2 separated events
 - absolute timing (second)
- *Pop Music Transformer: Beat-based Modeling and Generation of Expressive Pop Piano Compositions* (Yu-Siang Huang, Yi-Hsuan Yang)
 - **REMI** representation
 - note off -> duration
 - absolute timing -> symbolic timing (beat)

Representation: REMI

- duration

...

<note_on, note=60, velocity=94, time=0.0 sec>

<note_on, note=60, velocity=0, time=1.253 sec>

<note_on, note=64, velocity=94, time=1.253 sec>

<note_on, note=64, velocity=0, time=1.879 sec>

<note_on, note=67, velocity=94, time=1.879 sec>

<note_on, note=67, velocity=0, time=2.506 sec>

...

Sequence Length: 54
9 (notes) x 2 (note on/off) x 3 (attributes)

...

<note_on, note=60, **duraiton=4**, velocity=94, time=0 tick>

<note_on, note=64, **duraiton=2**, velocity=94, time=4 tick>

<note_on, note=67, **duraiton=2**, velocity=94, time=6 tick>

...

Sequence Length: 36
9 (notes) x 4 (attributes)

Representation: REMI

<note_on, note=60, duraiton=4 **tick**, velocity=94, time=0 **tick**>

- Symbolic Timing System

Note (pitch=60, velocity=72)



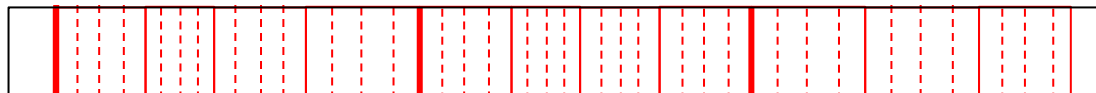
sec (or millisec)



beat / downbeat

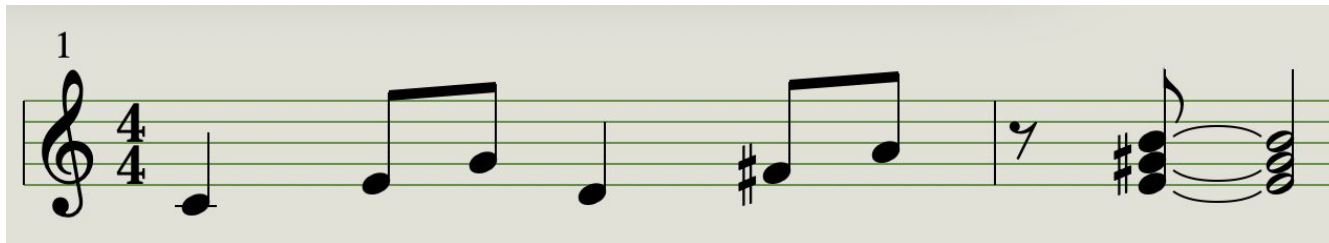
85 85 84 82 86 88 85 83 82 81 84 ...

bpm



tick (sub-beat)

Representation: REMI



<bar>

<tempo, bpm=85, tme=0 tick>

```
<note_on, note=60, duraiton=4, velocity=94, time=0 tick>
```

<tempo, bpm=85, tme=4 tick>

```
<note_on, note=64, duraiton=2, velocity=94, time=4 tick>
```

```
<note_on, note=67, duraiton=2, velocity=94, time=6 tick>
```

...

<bar>

...

Sequence Length: 54

**9 (notes) x 4 (attributes) +
8 (tempos) x 2 (attributes) +
2 (bars)**

Representation: REMI v2

- Remove duplicated tokens

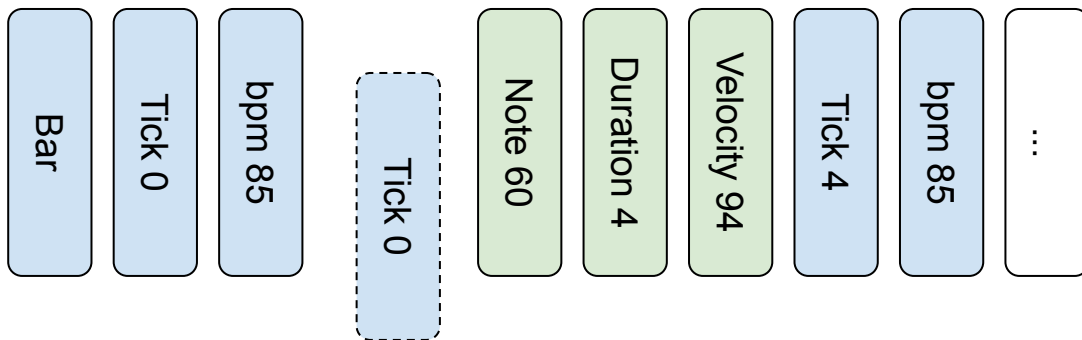
<bar>

<tempo, bpm=85, tme=0 tick>

<note_on, note=60, duraiton=4, velocity=94, time=0 tick>

<tempo, bpm=88, tme=4 tick>

...



Remove

Sequence Length: 50

**9 (notes) x 4 (attributes) +
8 (tempos) x 2 (attributes) +
2 (bars) -
4 (duplicated events)**

Representation: Compound Word (CP)

- E events, A attributes for each,
 - $O(EA)$ steps in total
- **grouping**
- **multi-headed output layer**

Sequence Length: 30

9 (notes) +
8 (tempos) +
2 (bars) +
11 (tick positions)

	head 1	head 2	head 3
step 1	Bar	<ignore>	
	Tick 0		
	bpm 85		
	Duration 4	Note 60	Velocity 94
	Tick 4		
	bpm 85		
step 7	Duration 2	Note 64	Velocity 94

Representation: CP

- expansion

	head 1	head 2	head 3	head 4	head 5	head 6
step 1	Bar					
step 2		Tick 0				
step 3			Bpm 85			
step 4				Duration 4	Note 60	Velocity 94

Representation: Recap

- From MIDI to CP
 - Symbolize timing
 - Reduce about half sequence length
- The reduction in sequence length would be even greater when we add more attributes per events (e.g. chord)

Model: Multi-headed Output

	head 0	head 1	head 2	head 3	head 4	head 5	head 6
step 1	Metrical	Bar					
step 2	Metrical		Tick 0	Bpm 85			
step 3	Note				Duration 4	Note 60	Velocity 94
step 4	Metrical		Tick 4	<conti>			
step 5	Note				Duration 2	Note 64	Velocity 94
	Family token	Metrical related token			Note related token		

Model: Multi-headed Output

Stage 1

$$\mathbf{h}_t = \text{Self-attn}(\vec{\mathbf{x}}_{t-1}) ,$$

$$\hat{f}_t = \text{Sample}_{\mathcal{F}}(\text{softmax}(\mathbf{W}_{\mathcal{F}}\mathbf{h}_t)) ,$$

Stage 2

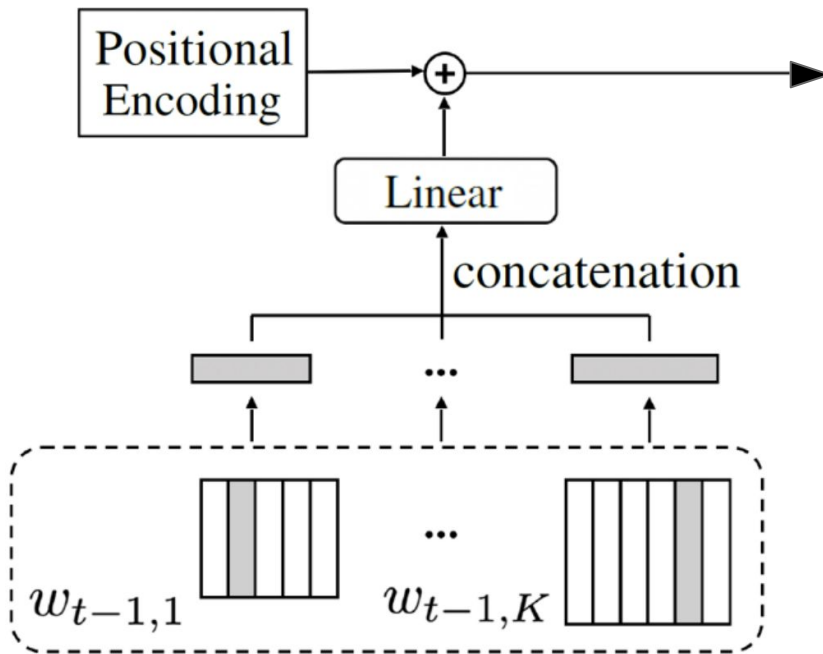
$$\mathbf{h}_t^{\text{out}} = \mathbf{W}_{\text{out}}[\mathbf{h}_t \oplus \text{Embedding}_{\mathcal{F}}(\hat{f}_t)] ,$$

$$\widehat{w_{t,k}} = \text{Sample}_k(\text{softmax}(\mathbf{W}_k\mathbf{h}_t^{\text{out}})) , \quad k = 1, \dots, K ,$$

- Training: teacher forcing
- Inference: stochastic sampling

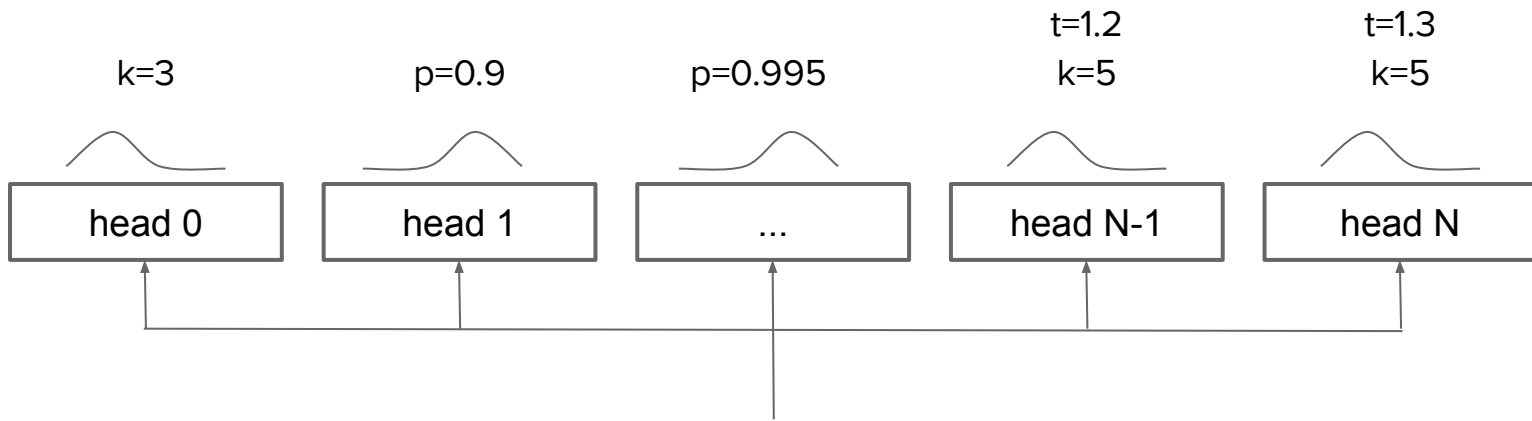
Model: Adaptive Embedding

- Difficulty of learning
 - Hard: velocity
 - Easy: bar & beat



Model: Adaptive Sampling

- Inference Stage
 - temperature t
 - top-k
 - top-p (nucleus)



Model: Compound Word Transformer

- **Compound Word Transformer (CP Transformer)**
 - Compound Word Representation
 - Adaptive Embedding
 - Multi-Headed Output Module
 - Two-stage Prediction
 - Adaptive Sampling (Inference Time)

Dataset

- Corpus
 - 17K pop piano music dataset
- Tasks
 - Unconditional Generation
 - Conditional Generation:
 - lead sheet to performance



Audio Clip



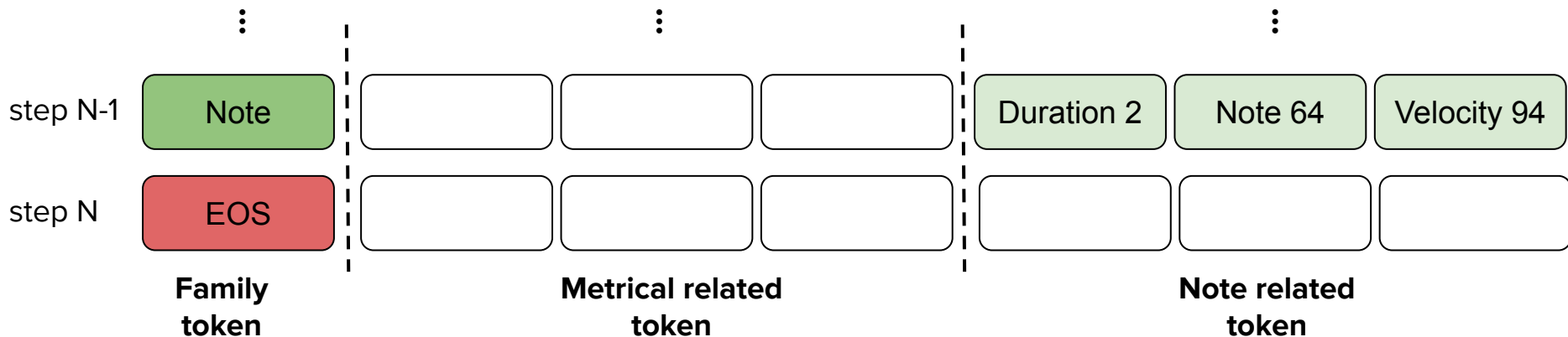
- Transcription: Onset and Frames
- Synchronization: madmom
- Quantization
- Analysis
 - Melody Extraction
 - Symbolic Chord Recognition



MIDI File

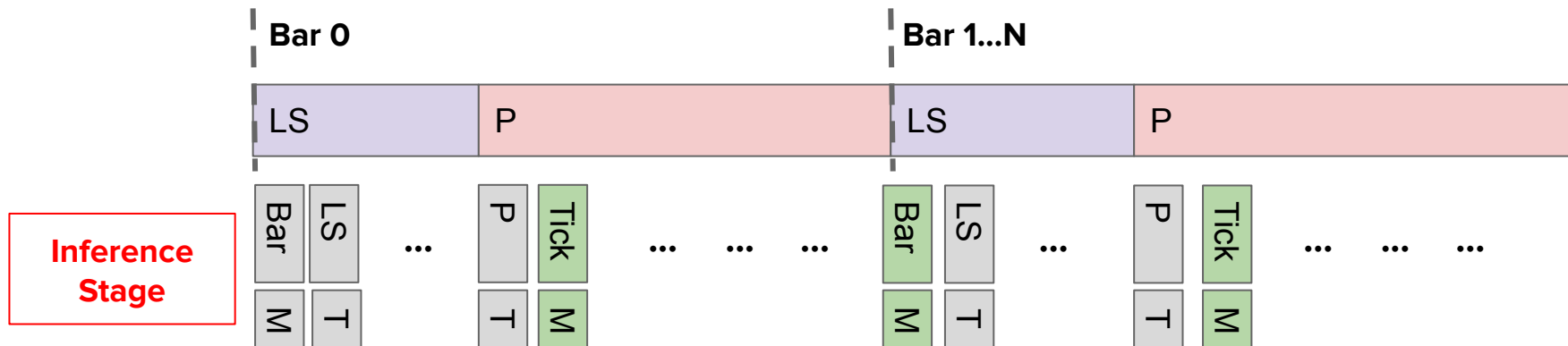
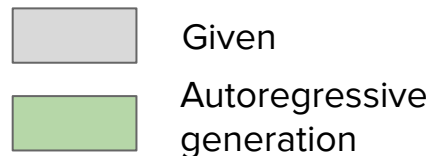
Task1: Unconditional Generation

- Beat Synchronized Feature
 - Chord
 - Tempo
- New type token: **End of sequence** <EOS>



Task2: Conditional Generation

- lead sheet to performance
- New type token: **track** (T)
 - Lead Sheet track (LS)
 - Piano track (P)
- Encoder-free prefix LM (from google's "T5")



Experiments

- **Sequence Length**

Task	Repre.	#words (T)	
		mean ($\pm$ std)	max
Conditional	REMI	6,432 ($\pm$ 1,689)	10,240
	CP	3,142 ($\pm$ 821)	5,120
Unconditional	REMI	4,873 ($\pm$ 1,311)	7,680
	CP	2,053 ($\pm$ 580)	3,584

- **Backbone Models**

- **Transformer-XL:** recurrence
- **Linear Transformer:** kernelization

Quantitative Evaluation

- Single GPU (2080ti, 11GB)
- Training & Inference Time

Task	Representation + model@loss	Training time	GPU memory	Inference (/song)	
				time (sec)	tokens (#)
Conditional	Training data	—	—	—	—
	Training data (randomized)	—	—	—	—
	REMI + XL@0.44	3 days	4 GB	88.4	4,782
	REMI + XL@0.27	7 days	4 GB	91.5	4,890
	REMI + linear@0.50	3 days	17 GB	48.9	4,327
	CP + linear@0.27	0.6 days	10 GB	29.2	18,200
Unconditional	REMI + XL@0.50	3 days	4 GB	139.9	7,680
	CP + linear@0.25	1.3 days	9.5 GB	19.8	9,546

Quantitative Evaluation

- Metrics for conditional generation
 - Melody matchness
 - Bar-wise Longest Common Sub-sequence (LCS) Matchness

$$Matchness_{Melody} = \frac{|LCS(Seq_{Melody}, Seq_{Piano})|}{|Seq_{Melody}|}$$

- Chord matchness
 - Segmentwise Cosine Similarity of chormagramss

Quantitative Evaluation

	melody Matchness	chord Matchness
Training data	0.755	0.838
Training data (randomized)	0.049	0.239
REMI + XL@0.44	0.872	0.785
REMI + XL@0.27	0.866	0.800
REMI + linear@0.50	0.779	0.709
CP + linear@0.27	0.829	0.733

User Study

Repre. + model@loss	F	R	H	C	O
REMI + XL@0.44	4.05	3.12	3.38	3.55	3.31
REMI + XL@0.27	4.29	3.14	3.70	3.64	3.35
REMI + linear@0.50	4.03	3.09	3.48	3.46	3.29
CP + linear@0.27	4.09	3.13	3.50	3.31	3.08

(a) Conditional generation

Repre. + model@loss	R	H	S	O
REMI + XL@0.50	3.11	3.46	2.91	3.03
CP + linear@0.22	3.33	3.68	3.11	3.34

(b) Unconditional generation

Table 5: Result of subjective evaluation (**F**idelity, **R**ichness, **H**umanness, **C**orrectness, **S**tructureness, **O**verall).

Summary

- Long sequence modeling
 - memory-efficient transformer
 - compact representation
- Compound Word Transformer
- Faster in training and inference, with comparable performance

Summary

- Full Song Generation?
 - EOS token generation
 - transformer-XL 
 - linear transformer 
 - Structural Pattern? (like AABA forms)
 - still No

Demo

- Sound Cloud
 - https://soundcloud.com/yating_ai/sets/compound-word-transformer-demo
- Github
 - <https://github.com/YatingMusic/compound-word-transformer>